

Jeongseo Jang

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Research Interests | *Wearable Robotics, Exoskeleton Control, Biosignal-based Human-Robot Interaction, Physics-based Simulation, Reinforcement Learning for Assistive Robotic Systems*

Education

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| Kumoh National Institute of Technology , Gumi, Republic of Korea | Mar 2024 – Aug 2026 |
| • M.S. Mechanical Engineering, expected Aug 2026 | |
| Kumoh National Institute of Technology , Gumi, Republic of Korea | Mar 2020 – Feb 2024 |
| • B.S. Mechanical Design Engineering | |

Research Experiences

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| Bio-Interfaced Translational Nanoengineering Group , Georgia Tech | Aug 2025 – Feb 2026 |
| <i>International Visiting Researcher. Supervisor: Prof. W. Hong Yeo</i> | |
| <ul style="list-style-type: none">Conducted research on AI-powered noise-robust EMG patches for dual robotic hand control, focusing on biosignal-driven human-robot interaction.Developed EMG-driven dexterous manipulation tasks in a virtual reality environment using reinforcement learning-based control.Integrated biosignal interpretation and robotic hand control to explore assistive and wearable robotic applications.Contributed to a manuscript on intelligent human-robot interaction enabled by advanced biosignal detection. | |

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| AI & Autonomous System Lab , Kumoh National Institute of Technology | Jan 2022 – Aug 2026 |
| <i>Undergraduate and M.S. Researcher. Supervisor: Prof. Jung Woo Sohn</i> | |
| <ul style="list-style-type: none">Built reinforcement learning-based robotic manipulation simulations for path planning and control policy optimization.Designed EMG-driven robotic manipulation frameworks integrating biosignal recognition, robot control, and virtual simulation.Developed ROS-based robot arm and autonomous mobile robot systems for smart factory applications, including motion recognition-based control and object recognition.Developed adaptive control algorithms for active vehicle suspension and analyzed control characteristics under dynamic conditions.Developed a reinforcement learning-based autonomous emergency braking system in simulation for autonomous vehicle safety control. | |

Publications

- Cheoljeong Park, **Jeongseo Jang**, Jung Woo Sohn, Seong J. Cho & Woon-Hong Yeo. Intelligent Human-Robot Interaction Enabled by Advanced Biosignal Detection. Advanced Robotics, under review.
- Jang, J. S.**, & Sohn, J. W. (2026). Reinforcement Learning-Based Vehicle Autonomous Emergency Braking System. Journal of the Korean Society of Manufacturing Process Engineers, 25(2), 48-56.
- Jang, J. S.**, & Sohn, J. W. (2024). Control Characteristics of Active Suspension in Vehicles using Adaptive Control Algorithm. Transactions of the Korean Society for Noise and Vibration Engineering, 34(5), 568-577. 10.5050/KSNVE.2024.34.5.568

4. **Jang, J. S.**, Park, M., Kim, M., Lee, J., & Sohn, J. W. (2023). User Motion Recognition Based Robot Arm Control Using ROS. *Journal of the Korean Society of Manufacturing Process Engineers*, 22(10), 29-35.

Conference Presentations

1. **J. S. Jang** & J. W. Sohn, "EMG-based Reinforcement Learning for Dexterous Grasping in Virtual Reality Environment", KSPE, Jeju, Korea (May. 2026) – Poster
2. **J. S. Jang** & J. W. Sohn, "Optimal Path Planning for a Robot Manipulator using Reinforcement Learning in Simulation", KSME, Gyeongju, Korea (Apr. 2026) – Poster
3. **J. S. Jang** & J. W. Sohn, "Optimal Path Planning for a Robot Arm using Reinforcement Learning", KSNVE, Jeju, Korea (May. 2025) – Oral
4. **J. S. Jang** & J. W. Sohn, "Autonomous Driving Simulation based on Reinforcement Learning", KSMPE, Yeosu, Korea (May. 2025) – Poster
5. **J. S. Jang** & J. W. Sohn, "Collision Avoidance System for Autonomous Vehicles based on Simulation", KSMPE, Jeju, Korea (Nov. 2024) – Oral
6. **J. S. Jang** & J. W. Sohn, "AI-based Simulation of Autonomous Driving", KSME, Jeju, Korea (Nov. 2024) – Poster
7. **J. S. Jang** & J. W. Sohn, "Active Control of Automobile Suspension using Adaptive Control", KSEM, Yeosu, Korea (Apr. 2024) – Oral
8. **J. S. Jang** & J. W. Sohn, "Autonomous Driving using Image-based Object Recognition System", KSMPE, Gwangju, Korea (Apr. 2024) – Poster
9. **J. S. Jang** & J. W. Sohn, "Tracking System using Object Recognition Algorithm", KSMPE, Jeju, Korea (Nov. 2023) – Poster
10. **J. S. Jang** & J. W. Sohn, "Image-Based Object Recognition for Autonomous Mobile Robot", International Conference on PRESM, Okinawa, Japan (Jul. 2023) – Poster
11. **J. S. Jang**, S. Y. Jeong, M. J. Park, J. W. Sohn, "Robot Motion Generation Using Human Motion Recognition", KSNVE, Jeju, Korea (Nov. 2022) – Poster

Selected Honors and Awards

Research Encouragement Grant for Outstanding Paper Presentation , <i>Korean Society for Noise and Vibration Engineering and NSV Co.</i>	<i>Oct 2024</i>
• Awarded for an outstanding conference paper presentation with a KRW 2,500,000 research grant	
Outstanding Research Paper Presentation Award , <i>Korean Society of Mechanical Engineers</i>	<i>Oct 2023</i>
• Recognized for an outstanding research presentation on continuous fiber-reinforced composite 3D printing for lightweight drone structures.	
Design and Idea Award , <i>Startup Club Competition, Kumoh National Institute of Technology</i>	<i>Jan 2022</i>
• Award for a startup-oriented engineering product concept: a real-time total noise measurement device	

Extracurricular Activities and Leadership

Student Executive Member , <i>Korea Society of Manufacturing Process Engineers</i>	<i>Nov 2024</i>
• Appointed to support academic and student activities within the society.	
Teaching Assistant, SW Reskilling Program: MATLAB Simulink , <i>HYUNDAI NGV</i>	<i>Jun 2024</i>
• Assisted participants with hands-on MATLAB/Simulink exercises and supported training sessions on modeling and simulation.	

Intellectual Property

Autonomous Carrier Moving with a User, Invention Disclosure,
submitted to the Kumoh National Institute of Technology

Aug 2024

- Co-inventor, 25% contribution

Skills Summary

Programming Languages: Python, C, C++, MATLAB

Machine Learning / RL: PyTorch, TensorFlow, Reinforcement Learning, Isaac Lab

Robotics / Simulation: ROS, Isaac Sim, IPG CarMaker, MATLAB/Simulink

CAD/CAE: AutoCAD (2D), Inventor (3D), ANSYS

Hardware: Raspberry Pi, Arduino

Languages: Korean (Native), English (Intermediate)